

Number theory – 6

Arithmetic functions and Dirichlet convolution

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Theoretical minimum. Arithmetic functions form a ring under convolution, and almost every identity of elementary number theory is an identity in that ring.

Definitions. An arithmetic function is a map $f: \mathbb{N} \rightarrow \mathbb{C}$; it is *multiplicative* if $f(mn) = f(m)f(n)$ for coprime m, n and $f(1) = 1$. The standard ones are φ , the number of divisors $d(n)$, their sum $\sigma(n)$, the Möbius function $\mu(n)$ — zero if n is not squarefree and $(-1)^k$ for a product of k distinct primes — and the constant **1**. The *Dirichlet convolution* is

$$(f * g)(n) = \sum_{d|n} f(d) g(n/d).$$

Theorem 1 (the convolution ring). Convolution is commutative, associative and distributive, with identity $\varepsilon(n) = [n = 1]$; f is invertible exactly when $f(1) \neq 0$; and the convolution of two multiplicative functions is multiplicative. *Proof* of the last part: for coprime m, n every divisor of mn factors uniquely as a divisor of m times a divisor of n .

Theorem 2 (Möbius inversion). $\mu * \mathbf{1} = \varepsilon$, that is $\sum_{d|n} \mu(d) = [n = 1]$; consequently $g = f * \mathbf{1}$ if and only if $f = g * \mu$, i.e.

$$g(n) = \sum_{d|n} f(d) \iff f(n) = \sum_{d|n} \mu(d) g(n/d).$$

Examples: $\mathbf{1} * \mathbf{1} = d$, $\text{id} * \mathbf{1} = \sigma$, $\varphi * \mathbf{1} = \text{id}$ — the last being $\sum_{d|n} \varphi(d) = n$, proved by grouping the fractions k/n by their reduced denominator — whence $\varphi = \mu * \text{id}$ and $\varphi(n) = n \prod_{p|n} (1 - 1/p)$.

Theorem 3 (Dirichlet series). Formally $\sum_n \frac{(f*g)(n)}{n^s} = \left(\sum_n \frac{f(n)}{n^s} \right) \left(\sum_n \frac{g(n)}{n^s} \right)$, so convolution becomes multiplication of series, and a multiplicative f has the

Euler product $\sum_n f(n)n^{-s} = \prod_p (1 + f(p)p^{-s} + f(p^2)p^{-2s} + \dots)$. In particular $\sum \mu(n)n^{-s} = 1/\zeta(s)$, $\sum d(n)n^{-s} = \zeta(s)^2$ and $\sum \varphi(n)n^{-s} = \zeta(s-1)/\zeta(s)$. Questions of convergence and the summation of such series belong to Series and infinite products – 5 and – 6 and are not repeated here.

Fact 1 (average orders). $\sum_{n \leq x} d(n) = x \ln x + (2\gamma - 1)x + O(\sqrt{x})$ by Dirichlet's hyperbola method; $\sum_{n \leq x} \varphi(n) = \frac{3}{\pi^2}x^2 + O(x \ln x)$, so the probability that two random integers are coprime is $6/\pi^2$; and $\sum_{n \leq x} \sigma(n) = \frac{\pi^2}{12}x^2 + O(x \ln x)$.

1. Prove that the arithmetic functions form a commutative ring under convolution, that a function with $f(1) \neq 0$ is invertible, and that the convolution of multiplicative functions is multiplicative. Then compute the convolution inverse of $\mathbf{1}$, of d and of σ .
2. Prove $\sum_{d|n} \varphi(d) = n$ in two ways — by grouping fractions and by multiplicativity — and deduce Möbius inversion together with the formula $\varphi(n) = n \prod_{p|n} (1 - 1/p)$. Then prove $\sum_{d|n} \mu(d)d(n/d) = 1$ and $\sum_{d|n} \mu(d)\sigma(n/d) = n$.
3. Find all natural numbers n such that $n \sigma(n) \equiv 2 \pmod{\varphi(n)}$. (IMS 2008)
4. Prove the hyperbola method and deduce $\sum_{n \leq x} d(n) = x \ln x + (2\gamma - 1)x + O(\sqrt{x})$. Then prove that the probability that two integers chosen at random from $\{1, \dots, N\}$ are coprime tends to $6/\pi^2$, and identify where the value of $\zeta(2)$ enters.
5. Prove that n is even and perfect if and only if $n = 2^{k-1}(2^k - 1)$ with $2^k - 1$ prime (Euclid and Euler), and prove that an odd perfect number, if one exists, is not a prime power and has at least three distinct prime divisors.